



# ANTU ROY

## Power Electronics Hardware & Control Engineer

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[Portfolio Website](#)

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### PROFILE

*Passionate Power Electronics Hardware Engineer (R&D), Experienced in the Design, Development, and Validation of Power Converter Systems. Working on N+1 Redundant UPS Power Modules for Mission-Critical Data Center Applications, with Expertise in Power Stage Design, Gate Driver Circuits, Sensing Circuits, Protection Circuits, and System Integration. Experienced in Programming TI C2000 Microcontrollers, Including the TMS320F28379D, with Expertise in **PWM Generation**, **ADC Integration**, and MCU Peripheral Configuration. Skilled in Advanced Digital Control Systems Including PI/PR Controllers, 2P2Z and 3P3Z Compensators, **FOC**, **SPWM**, **Two-Level SVPWM**, Three-Level SVPWM, and PLL Synchronization. Proficient in MATLAB/Simulink, PLECS, Embedded C, Code Composer Studio (CCS), **Altium Designer**, LTspice, and PSIM, with Hardware Bring-Up, Debugging, and Prototype Testing Experience.*

### EXPERIENCE

#### **AMETEK Instruments India Pvt. Ltd.**

**Nov 2025 – Present**

*Power Electronics Hardware Engineer (R&D)*

- *Developing a 25kVA N+1 Redundant UPS Power Module for Data Center Applications, including Power Stage Design, Control Implementation, Protection Circuits, and Prototype Realization.*
- *Performing Engineering Analysis, Component Evaluation, Design Reviews, and Electrical Validation for High-Power Converter Systems.*
- *Conducting Prototype Testing, Hardware Debugging, Root-Cause Analysis, and Corrective Actions during Product Validation Cycles.*
- *Collaborating with Cross-Functional Teams for Design Optimization, Testing Activities, and Reliability Improvement of Power Conversion Systems.*

### TECHNICAL STRENGTHS

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|---------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Programming &amp; Tools:</b> | <i>Embedded C, CCS, MATLAB/Simulink, Model-Based Code Generation (MBD) and Git</i>                |
| <b>Peripherals:</b>             | <i>ePWM, ADC (12 &amp; 16-Bit), DAC, Timers &amp; Real-Time ISR &amp; Interrupt-Based Control</i> |
| <b>Control &amp; Modeling:</b>  | <i>PI and PR Controllers and D-Q Control, SOGI-PLL and MPPT Algorithms Techniques</i>             |
| <b>Circuit Design:</b>          | <i>LTspice, PLECS, PSpice, TINA-TI, SIMetrix, PSIM and Proteus Simulation Platforms</i>           |
| <b>Components Selection:</b>    | <i>MOSFETs and IGBTs, Diodes, <b>Gate Drivers</b>, Snubber Circuits and Magnetic Components</i>   |
| <b>Hardware &amp; Testing:</b>  | <i>Oscilloscopes, Power Analyzers, Debugging, Fault Analysis, Tools, Prototyping and Setup</i>    |
| <b>Domains:</b>                 | <i>Inverters and Converters, Motor Drives, PV Systems and Grid Integration Focus Areas</i>        |
| <b>Document Creation:</b>       | <i>MS Word and MS Excel, MS Visio, LaTeX, Google Docs, Overleaf and Adobe Photoshop</i>           |

### EDUCATION

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|----------------------------------------------------------------------------|--------------------------|
| <b>National Institute of Technology Calicut</b> , Kozhikode, Kerala        | <b>2023-2025</b>         |
| <i>Master of Technology in Power Electronics</i>                           | <i>CGPA: 7.77</i>        |
| <b>Regent Education and Research Foundation</b> , Barrackpore, West Bengal | <b>2016-2020</b>         |
| <i>Bachelor of Technology in Electrical Engineering</i>                    | <i>CGPA: 7.58</i>        |
| <b>Asannagar High School (H.S)</b> , Krishnagar, Nadia, West Bengal        | <b>2014-2016</b>         |
| <i>West Bengal Council of Higher Secondary Education</i>                   | <i>Percentage: 71.20</i> |
| <b>Asannagar High School (H.S)</b> , Krishnagar, Nadia, West Bengal        | <b>2012-2014</b>         |
| <i>West Bengal Board of Secondary Education</i>                            | <i>Percentage: 80.14</i> |

## HARDWARE EXPERIENCE

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### Advanced Power Electronics

- *Designed and Developed Buck, Boost & Buck-Boost Converters for Reliable High-Efficiency Power Conversion*
- *Developed 1-Phase and 3-Phase Inverters Using SPWM Technique to Voltage Control and Output Performance*

### Digital Control and Embedded Systems

- *Embedded Firmware: ISR Design, Fixed-Point Math, Lookup Tables, Calibration, Logging Over UART/UART*
- *MCU Peripherals: ADC Trigger Chains, ePWM Sync, Deadtime, Complementary Outputs, DAC and GPIO*

## NOTABLE HARDWARE PROJECTS

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### M.Tech Projects

#### **Design & Control of Single-Stage Grid-Connected Inverter for 1- $\phi$ PV System** | CCS Monsoon 2025

- *Designed and Simulated Single-Stage Inverter in PLECS/MATLAB with MPPT and Grid Sync on TI C2000*
- *Optimized Control on TI C2000 for High Efficiency and Validated Performance Across Diverse Grid Scenarios*

#### **Field-Oriented Control of Three-Phase Induction Motor Using SVPWM Technique** | CCS Winter 2025

- *Built and Simulated FOC-Based Control for 3- $\phi$  ACIM in MATLAB/Simulink Using SVPWM on TI C2000*
- *Optimized Control on TI C2000 for Exact Speed, High Efficiency & Stable Dynamics Across Load Conditions*

#### **Design & Analysis of 3- $\phi$ Inverter Using Sinusoidal ePWM Modulation Technique** | CCS Winter 2024

- *Designed and Simulated Open-Loop 3- $\phi$  Inverter in PLECS/MATLAB & Implemented Hardware on TI C2000*
- *Optimized ePWM Control on TI C2000 for Efficient Sinusoidal Output and Evaluated Inverter Performance*

#### **Control of DC Motor Using ePWM on the TMS320F28379D Microcontroller** | CCS Monsoon 2023

- *Designed and Optimized PWM Control Algorithms for Precise Motor Speed Control and Efficient Performance*
- *TMS320F28379D is Utilized to Implement Precise Control by Modulating the Duty Cycle of the PWM Signals*

### B.Tech Projects

#### **Optimal Operational Scheduling of a Grid Connected System Based on DSM** | MATLAB Winter 2020

- *Solved the Demand Side Management Problem by Considering of a Grid Connected Network by Load Shifting*
- *Reduced the Supply Side's Cost and Increased the Demand Side's Revenue in a Grid Using the DE Algorithm*

## AREAS OF INTEREST

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- *Design and Control of Inverters for Photovoltaic (PV) Systems, PLL Grid Integration, and MPPT Algorithms*
- *Design and Control of DC-DC Converters, Advanced Control Techniques, and Power Converter Optimization*
- *Embedded Systems for Power Electronics Projects, Including TMS320F28379D Microcontroller Programming*
- *Electric Vehicle Powertrain Management, Battery Systems, Charging Infrastructure, and Energy Efficiency*

## TRAINING AND CERTIFICATES

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### Vocational Training Program

#### **West Bengal State Electricity Transmission Company Limited (WBSETCL)** Winter 2020

- *Trainee was Acquainted with the Electrical Operation & Maintenance and Functioning of Different Equipment Like Power Transformer, Current Transformer, Potential Transformer, Circuit Breaker, Lightning Arrestor Productive Relays, Conductors, Isolators, Transmission Towers, Electrical Safety Procedure & Devices, etc.*

#### **West Bengal State Electricity Distribution Company Limited (WBSEDCL)** Winter 2020

- *Electrical Operation & Maintenance Activities of Distribution Line & Sub Station Upto 33 kV of WBSEDCL*